

Wirebrowser

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README.md

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Wirebrowser

[[image: Open Source](#)] [[image: Built on CDP](#)] [[image: License MIT](#)] [[image: Contributions welcome](#)]

Wirebrowser is a **runtime instrumentation platform for JavaScript the browser**, built on top of the Chrome DevTools Protocol (CDP).

It brings some **Frida-like capabilities** to Chrome — hooking functions, inspecting runtime state, search memory, and modifying behavior without monkeypatching.

Wirebrowser lets you **observe, intercept, and modify execution at runtime**, even inside closures and non-global scopes that are normally unreachable.

It is designed for:

- reverse engineers
- security researchers / bug hunters
- advanced frontend developers

Core capabilities include:

- **Hooks** — inject logic and override behavior at runtime
- **Origin Trace (BDHS)** — automatically trace where a value is created or mutated
- **Live Object Search** — find and patch runtime objects
- **Network interception & replay** — modify inputs and observe effects

Unlike traditional tools, Wirebrowser focuses on **causality and execution flow**, not just inspection.

📄 Quick Links

- 📄 Documentation → <https://fcavallarin.github.io/wirebrowser/api/>
- 📄 API Documentation → <https://fcavallarin.github.io/wirebrowser/>
- ⚙️ CDP as a Runtime Instrumentation Engine Writeup → <https://fcavallarin.github.io/wirebrowser/CDP-as-a-Runtime-Instrumentation-Engine>
- 📄 BDHS / Origin Trace Writeup → <https://fcavallarin.github.io/wirebrowser/BDHS-Origin-Trace>
- ▶️ BDHS YouTube Demo → <https://www.youtube.com/watch?v=WA5nHk-6UJc>

📄 Overview

Wirebrowser is built around one core idea:

Move from inspection → to runtime control

⚡ Runtime Instrumentation

Hook functions at runtime using CDP breakpoints — without modifying source code.

- Inject logic during execution
- Override return values
- Observe arguments and behavior
- Instrument code inside closures (not reachable from `window`)

No monkeypatching. No fragile overrides.

📄 Memory & Causality Analysis

Understand not just *what exists*, but **where it comes from**.

- **Origin Trace (BDHS)** — identify the exact function responsible for creating or mutating a value ([writeup](#))
- **Live Object Search** — find and patch runtime objects
- **Heap Snapshot Search** — search full V8 memory (including unreachable values)

This bridges the **causality gap** in traditional debugging.

□ Network → Runtime Bridge

Intercept and modify inputs, then observe their effects in runtime:

- Rewrite HTTP responses
 - Replay requests
 - Correlate network data with runtime objects
-

□ Key Idea

Traditional tools answer:

“What is happening?”

Wirebrowser answers:

“Where did this come from, and how can I change it?”

□ Feature Highlights

Below is a quick visual tour of Wirebrowser’s most distinctive capabilities.

A short walkthrough of Wirebrowser’s advanced memory-analysis capabilities:

- **Live Object Search** — real-time search and runtime patching of live JS objects.
 - **Origin Trace (BDHS)** — identify the user-land function responsible for creating or mutating the object during debugging.
 - **Live Hooks** — hook the target function at runtime and override its return value or inject code.
-

Hooks - Runtime Instrumentation

Hook functions at runtime using CDP breakpoints — without modifying source code.

- Inject custom logic during execution
- Override return values
- Inspect arguments and behavior
- Works even inside closures (not reachable from `window`)

[\[image: Hooks\]](#)

Memory — Origin Trace (BDHS)

Automatically identify the function responsible for creating or mutating a value.

- Snapshot taken at each debugger pause
- Each snapshot is searched
- Framework/vendor code filtered via heuristics
- Includes tolerance window for contextual analysis

[\[image: Origin Trace\]](#)

Memory — Live Object Search

Search and **patch** live JS objects using regex or structural matching.

[\[image: Live Objects\]](#)

Network Interceptor

Intercept, rewrite, block, and replay HTTP requests and responses.

[\[image: Network Interceptor\]](#)

Getting Started

📦 Desktop Builds (Recommended)

Pre-built desktop applications are available for:

- macOS (.dmg)
- Windows (.exe)
- Linux (.AppImage)

You can download the latest builds from the GitHub Releases page:

📄 <https://github.com/fcavallarin/wirebrowser/releases>

📦 Build from Source

If you prefer to run Wirebrowser from source:

```
git clone https://github.com/fcavallarin/wirebrowser.git
cd wirebrowser
npm install
npm run build
```

Run

```
npm run wirebrowser
```

❏ Linux Notes

Sandbox issue with Electron

On some Linux distributions, Electron may fail to start due to process sandboxing restrictions, showing errors such as:

```
The SUID sandbox helper binary was found, but is not configured correctly.
```

This is a known issue in Electron ([https://github.com/electron/electron/issues/42510]). The most common solution is to disable AppArmor restrictions:

```
sudo sysctl -w kernel.apparmor_restrict_unprivileged_userns=0
```

Chrome extension location

When running Wirebrowser on Linux via the AppImage, Chromium security policies require unpacked extensions to be stored in a visible (non-hidden) directory.

For this reason, the Wirebrowser Chrome extension is installed in:

```
~/wirebrowser/chrome-extension
```

This behavior is intentional and required for Chromium to load the extension correctly.

⚠ Do not move, rename, or hide this directory, otherwise the extension will fail to load.

► Scope of actions — Global vs Tab-specific

Most Wirebrowser actions can be performed **either globally (across all open tabs/pages)** or **targeted to a single tab**. This lets you choose whether a rule or inspection should affect the whole browser session or only a specific page. Every tab/page opened by Wirebrowser has a unique integer `tabId`. Use this `tabId` to scope actions.

UI Notes

- Many panels offer a **scope selector** (Global / Specific Tab ID) for quick changes.

❏ Contributing

Contributions and pull requests are welcome! Open an issue or pull request — even small suggestions help improve Wirebrowser.

❏ License

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layout: default title: Breakpoint-Driven Heap Search (BDHS)

Tracing JavaScript Value Origins with Breakpoint-Driven Heap Search (BDHS) and Live Object Search

Modern JavaScript applications make a deceptively simple question extremely hard to answer:

Where does this value come from?

Between frameworks, bundlers, minifiers, async flows, closures, and virtual DOM abstractions, the origin and lifecycle of a value become opaque. You might see:

- a token inside a request header,
- a corrupted component state,
- a tainted DOM string,
- a suspicious flag affecting behavior,

...with no visibility into where it was created or modified.

Traditional debugging tools weren't designed for this reality. Manual breakpoints, stepping, logs, or a single heap snapshot rarely help.

This writeup presents the techniques implemented in [Wirebrowser](#), an open-source CDP-based toolkit:

- **Breakpoint-Driven Heap Search (BDHS / Origin Trace)** – a multi-snapshot temporal analysis technique for finding where objects *and primitives* first appear.
- **Live Object Search** – a runtime heap explorer capable of scanning and patching live objects.
- **Hybrid Structural Similarity Search** – a similarity engine that compares objects by structure and shape, available across *all* memory subsystems.

1. Why Tracing Value Origins Is Hard in Modern JavaScript

Several factors obscure causality:

- **Framework abstractions**

React, Vue, Angular, Svelte, and others wrap user-land logic and hide real callsites.

- **Bundlers and minifiers**

Applications ship as giant anonymous bundles; stack traces are meaningless.

- **Async flows**

Promises, microtasks, events, timers, schedulers — traditional linear stepping cannot follow these cross-task transitions.

- **Closures and hidden state**

Values can live inside unreachable closures or internal structures.

- **Static heap snapshots lack temporal information**

A snapshot shows what exists *now*, not where or when it appeared.

Even trivial questions like “*Where was this object created?*” become difficult.

2. Breakpoint-Driven Heap Search (BDHS / Origin Trace)

BDHS turns heap snapshots into a temporal search space.

Instead of capturing one snapshot at an arbitrary moment, BDHS:

1. Pauses execution automatically at meaningful boundaries.
2. Captures a heap snapshot at each pause.
3. Searches every snapshot for the target value or structure.
4. Identifies the **first snapshot** containing it (origin in time).
5. (Future) Detects when the structure or value changes across snapshots.
6. Maps the origin back to the **user-land function** responsible, ignoring framework noise.

BDHS answers:

“When did this value first appear, and which user-land function introduced it?”

Even inside minified, async-heavy SPAs.

2.1 Event → Heuristic Handler Detection → Step-Out Execution

BDHS does **not** pause randomly or on every async boundary. Instead, it follows this sequence:

1. Install an initial breakpoint on a *real user event* (e.g., click).

2. Attempt to identify the **actual event handler** for the event target using heuristics (*Handler Detection*). This is not fully deterministic but works in most real-world SPAs.

1. Attach a breakpoint to the resolved handler.
2. Execute with **step-out stepping**, pausing only at function returns.
3. Capture a snapshot at each pause.

This yields a clean, semantically meaningful timeline aligned with:

- real UX-driven control flow,
- user-land logic boundaries,
- places where values are typically created or transformed.

Unlike traditional linear stepping, BDHS does not attempt to follow every instruction or every async boundary. Instead, it uses semantic step-out execution: it pauses only at meaningful user-land function boundaries, creating a sparse but highly informative temporal timeline.

Even though BDHS issues step-out commands sequentially, it does not produce a linear execution trace. Traditional stepping tries to follow the real instruction flow, including async jumps and framework internals. BDHS instead samples only meaningful user-land boundaries, creating a sparse, semantic timeline rather than a faithful step-by-step trace.

2.2 Snapshot Capture & Search

At each pause, BDHS calls `HeapProfiler.takeHeapSnapshot`.

Snapshots include:

- all objects,
- all primitives (including strings),
- closure-captured values,
- internal framework data structures,
- unreachable nodes.

BDHS searches each snapshot for:

- object identity,
- nested keys/values,
- strings/primitives,
- structural shapes,
- **hybrid structural similarity**.

Since snapshots contain *everything*, BDHS can locate:

- closure-bound values DevTools cannot reach,
- ephemeral objects that live briefly,
- deeply nested cache entries,

- primitives unreachable from the global scope.
-

2.3 Identifying Origins

When a match is found:

- the **first snapshot** containing the object/value = its **origin**,
- (future) later snapshots that differ structurally = *mutation points*.

BDHS maps these events to the **user-land function** via:

- script URLs,
- byte offsets,
- framework/vendor blackboxing.

BDHS identifies the *function* where the value was introduced— not the exact line, due to CDP granularity and snapshot timing. BDHS can also provide the snapshots immediately before and after the first match through the tolerance window, offering contextual insight into how the value is created or mutated. BDHS therefore complements both static snapshot search and live object search: it provides the temporal dimension that neither of them can offer.

2.4 Tolerance Window (Snapshot Sampling Window)

The *tolerance window* expands BDHS by providing not only the snapshot where the target value first appears, but also the immediate execution context around it.

When BDHS detects the first match, it automatically:

- samples a configurable number of snapshots before the match (e.g., 5), and
- a configurable number of snapshots after the match (e.g., 15).

Wirebrowser displays these snapshots in a chronological table, highlighting:

- steps where the value does not yet exist,
- the exact snapshot where it first appears,
- subsequent steps that show how the object evolves.

This contextual timeline makes it possible to understand:

- what functions prepare or influence the creation,
- whether precursor data structures were set up earlier,
- how the value mutates immediately after being introduced,
- whether multiple functions participate in its lifecycle.

The tolerance window dramatically increases BDHS's diagnostic power, especially in asynchronous, framework-heavy SPAs where objects may be assembled over several steps.

3. Hybrid Structural Similarity Search

Wirebrowser includes a hybrid similarity engine for comparing JavaScript objects by structure and shape.

Pure SimHash performs poorly on small or sparse objects. To address this, Wirebrowser uses a hybrid similarity metric combining:

- **Jaccard similarity** between structural tokens
- **fuzzy similarity** between nested keys

The result is robust similarity detection for:

- small objects,
- deeply nested structures,
- state objects,
- request/response payloads,
- objects that vary slightly between runs.

Similarity search is fully integrated across:

✓ **Live Object Search**

Find structurally similar objects in the **live heap**.

✓ **Static Heap Snapshot Search**

Identify clusters of similar objects inside snapshots.

✓ **BDHS (Origin Trace)**

Use structural similarity as a search key across temporal snapshots.

This enables queries like:

- “Find all objects shaped like this one.”
- “Trace the first appearance of this object family.”
- “Locate objects structurally similar to this payload.”

Because the same similarity engine runs across live memory, snapshots, and BDHS timelines, investigations can seamlessly pivot between the three without changing query semantics.

4. Static Snapshot Search vs BDHS vs Live Objects

Wirebrowser offers three memory analysis modes:

Static Heap Snapshot Search

- Captures a single snapshot.
 - Finds objects *and primitives*.
 - Read-only.
 - Useful to inspect a single moment in time.
-

BDHS / Origin Trace

- Captures *multiple* snapshots over time.
 - Tracks first appearance and (future) structural deltas.
 - Maps origins to the user-land function.
 - Works with objects and primitives.
-

Live Object Search

- Scans the **live heap** via CDP.
 - Finds objects by keys, values, structure, or similarity.
 - **Supports runtime patching** (mutating live objects).
 - Cannot detect primitives (V8 constraint).
-

5. Live Object Search and Runtime Patching

Live Object Search enables:

- finding objects unreachable from globals,
- scanning nested structures,
- pattern and similarity-based matching.

And supports **live object patching**:

- modifying fields,
- toggling flags,
- replacing handlers or callbacks,
- altering state/config values,
- injecting instrumentation.

Useful for:

- reproducing bugs without UI interaction,
- exploring edge cases,
- bypassing internal checks,
- security investigations,

- manipulating state machines in-flight.

6. Combined Workflow

The intended analysis workflow:

1. **Find** a suspicious value (Live Object Search).
 2. **Trace** its origin (BDHS / function-level origin).
- 2b. Use the **tolerance window** to inspect the snapshots **before** and **after** the origin, gaining contextual understanding of the value's creation or mutation.
1. **Patch** related objects to test hypotheses (Live Object Search).
 2. **Validate** by rerunning the scenario.

A cycle that traditional debugging cannot achieve:

find → trace → patch → validate

7. Concrete Examples

These scenarios show how BDHS, Live Object Search, and Hybrid Similarity Search work together in real investigations.

Example 1 — Tracing a JWT-like Value in a Request

A token-like string appears in an outgoing request. It doesn't exist in globals, and DevTools shows only framework internals.

Workflow

1. Trigger the UI action that generates the request.
2. Use Live Object Search (regex) to find the value in memory.
3. Reload the page to remove the value from memory.
4. Start BDHS using the found value.
5. BDHS performs heuristic handler detection, step-out stepping, and snapshot capture/search.
6. The first snapshot containing the token maps to the **user-land function** assembling it.

Result

BDHS reveals logic hidden inside minified SPA bundles.

Example 2 — Finding Where a React Component State Becomes Invalid

A component occasionally enters an impossible state (`loading: true` + `error: true`).

Workflow

1. Trigger the problematic UI action once.
2. BDHS:
 - attempts heuristic handler detection
 - executes step-out stepping
 - captures snapshots at each boundary
3. BDHS searches snapshot-by-snapshot using patterns and similarity.
4. The earliest snapshot with the invalid structure reveals the **responsible reducer function**.

Result

Something stepping cannot catch becomes visible immediately. With the tolerance window, BDHS also reveals the snapshots preceding the invalid state, making it clear which function prepares the inconsistent data before the reducer sets the final state.

Example 3 — Tracing a Tainted String in a DOM-Based XSS Case

A dangerous string appears in a DOM sink.

Workflow

1. Copy the tainted string.
2. Use Live Object Search to find all occurrences in memory.
3. Run BDHS on one instance.
4. BDHS identifies when the string first appears in snapshots.
5. The origin maps to the **user-land function** that introduced it.

Result

Effective for DOM XSS incident response.

Example 4 — Reverse Engineering a Minified SPA

A hidden boolean flag controls anti-bot behavior.

Workflow

1. Use Live Object Search to locate the flag and similar objects.
2. Start BDHS.
3. BDHS traces the first appearance of the flag.
4. Mapping reveals a small region in a huge minified file.

Result

The investigation narrows from ~200k lines to ~20.

Example 5 — Using Structural Similarity to Trace Complex Objects

A component generates slightly varying nested configuration objects.

Workflow

1. Locate an instance via Live Object Search.
2. Use Similarity Search to find structurally similar objects.
3. Run BDHS using the similarity signature as a search key.
4. The first snapshot containing any object of that “shape family” reveals the origin.

Result

BDHS + similarity find origins even for evolving object families.

Example 6 — Tracing Runtime-Generated Code Paths

Some SPAs generate handlers dynamically with closures.

Workflow

1. Find the handler via Live Object Search (by pattern or similarity).
2. Run BDHS on the function object.
3. BDHS finds the first snapshot containing the function (including closure environments).
4. Mapping identifies the **factory function** responsible.

Result

BDHS handles what DevTools cannot.

8. Future Work

Building on the current foundation:

- **Object Lifecycle Diffing**

Compare similarity across BDHS snapshots to visualize evolution.

- **Mutation Tracking in Snapshots**

Detect meaningful structural changes over time.

- **Flow Debugger**

A temporal visualization combining BDHS, snapshots, and similarity deltas.

- **Cross-Modal Correlation**

Correlate network payloads with memory objects via similarity.

- **Automated SPA Crawling**

Explore SPAs while attaching BDHS and Live Object Search to each action.

- **Deeper Framework Blackboxing**

More robust heuristics for ignoring framework internals in single-file bundles.

- **Adaptive tolerance window**

Heuristics that automatically adjust the sampling range based on object complexity or execution patterns.

9. Closing Notes

Breakpoint-Driven Heap Search, Hybrid Similarity Search, and Live Object Search provide a new way to analyze JavaScript:

- automatic pauses instead of manual breakpoints,
- temporal heap search instead of static snapshots,
- function-level origin tracing instead of stack inspection,
- full-heap live search instead of globals-only visibility,
- runtime patching instead of passive observation.

Wirebrowser makes these techniques practical in real-world, framework-heavy, async-intensive applications.

Contributions and feedback are welcome.

10. Resources

- **Wirebrowser Repository**

<https://github.com/fcavallarin/wirebrowser>

- **BDHS + Live Object Search Demo Video**

<https://www.youtube.com/watch?v=WA5nHk-6UJc>

- **Documentation / Writeups**

This document is part of the Wirebrowser technical documentation inside the `docs/` directory.

[View original document](#)

CDP as a Runtime Instrumentation Engine

Hooks, Stepping, and Following Async Execution

This is not a wrapper around DevTools.

It is a different execution model built on top of the same primitives.

No monkeypatching. No proxies. No rewriting application code. Only debugger-level control over execution.

Modern web applications have become difficult to reason about at runtime.

When reverse engineering such applications — whether for security research, pentesting, or debugging — the main questions usually are:

- Where does this value come from?
- How can I find and patch a live object in memory?
- Where does this value go next?
- How can I hook a function without modifying the runtime?

The first problem can be addressed with techniques like Breakpoint-Driven Heap Search (BDHS), which focuses on identifying where a value is created. The second problem can be solved with the `Runtime.queryObject` + `Runtime.callFunctionOn` primitives of CDP.

Both techniques are analysed in detail in a previous writeup:

→ <https://fcavallarín.github.io/wirebrowser/BDHS-Origin-Trace>

That article focuses on:

- identifying where a value is created (Origin Trace / BDHS)
- locating live objects in memory (`Runtime.queryObject`) and patch them (`Runtime.callFunctionOn`)

In this article, we focus on the next steps:

Where does this value go next?

How to hook a function without monkeypatching

The Problem with Traditional Debugging

Most debuggers follow a human-driven model:

```
breakpoint
↓
program pauses
↓
developer inspects state
↓
step / continue
```

This works, but it is slow, manual, and does not scale.

Modern JavaScript applications often involve:

- asynchronous flows (`await` , Promises)
- deeply nested callbacks
- dynamic object construction
- obfuscated or minified code
- frameworks generating large amounts of boilerplate

A typical workflow becomes:

- set a breakpoint
- step manually
- try to understand propagation
- repeat

There is a gap between the mental model of execution and what debuggers actually expose.

Debuggers show state. They do not show causality.

Imagine if you can follow the same model but with automation:

```
Identify a function
↓
Hook the entry point and the return points
↓
Inspect and modify variables
↓
Apply conditional stepping (if var1==true then stepOut())
↓
Follow the execution after a return point (even in async)
↓
Search the live memory for a specific object
↓
Repeat
```

This would turn the debugging into a programmable engine to instrument/observe/modify the execution flow.

Hooks Alone Are Not Enough

Hooking improves observability and patchability, but limited to the current function.

Consider this:

```
hook(target, {  
  onEnter(args) {},  
  onLeave(retval) {}  
})
```

This gives visibility into inputs and outputs, but still leaves a critical gap:

Who consumes the result next?

Answering this usually requires:

- inspecting the call stack
- identifying potential callers
- adding more hooks
- repeating

This remains a manual process.

What we actually want is something closer to **flow navigation**.

Why Hooking Fails in Modern JavaScript

Hooking works well in linear, synchronous code.

Modern JavaScript is neither.

Consider:

- async/await splitting execution into multiple microtasks
- Promise chains that decouple producer and consumer
- frameworks introducing indirection layers
- closures hiding execution context

As a result:

The consumer of a value is often not in the same call stack.

This breaks the traditional hook model.

Even with perfect hooks, you still need to manually reconstruct:

- where execution resumes
- which branch is taken
- which function consumes the result

This is where traditional hooking reaches its limit.

The Async Return Value Problem

Before describing the implementation, it is worth addressing a fundamental constraint that shaped the design.

CDP exposes `Debugger.setReturnValue`, an experimental primitive that overrides the return value of a function. It can only be called when execution is paused at a return point — not at arbitrary breakpoints.

In synchronous code, this works as expected. The caller consumes the return value directly from the active call frame, so overriding it there is sufficient.

In async functions that cross an `await` boundary, the situation is different. The frame paused at the return site belongs to a resumed continuation, while the consumer observes the final resolution of the function's Promise. Modifying the visible return value in the current frame does not necessarily modify the value ultimately observed by the consumer.

Critically, `Debugger.setReturnValue` does not fail in this case. The modification is applied — it simply does not propagate to where it needs to go. There is no error, no warning. The value appears changed at the return site and has no effect downstream.

For async functions, you cannot reliably patch a value at the producer. You have to reach the consumer.

This constraint directly motivated the design of `followReturn`. Rather than intercepting the return value, the approach is to follow the async continuation into the consumer's execution frame, where the value is already a live variable in scope. Patching happens there instead.

```
// Patching at the producer – silently ineffective for async:
onLeave(ctx) {
  ctx.return(modifiedValue) // applied but does not reach the consumer
}

// followReturn reaches the consumer instead:
onLeave(ctx) {
  ctx.followReturn()
}
onReturnFollowed(ctx) {
  ctx.setVariable("token", modifiedValue) // value is live here
}
```

This reframes the limitation as a design constraint: rather than fighting the runtime, the model works with it.

A Different Model: Event-Driven Debugging

Instead of treating debugging as a manual process, we can model it as a stream of runtime events:

```
runtime event
↓
user handler
↓
debugger action
```

Events include:

- function entry
- function return
- step information
- async continuation

Each event triggers a handler that can:

- inspect runtime state
- emit messages
- modify execution
- request further debugger actions

This turns the debugger into a **programmable execution engine**.

Hooks on Top of the Chrome DevTools Protocol

Modern browsers expose a powerful debugging interface via the Chrome DevTools Protocol (CDP), including:

- breakpoints
- runtime evaluation
- async stepping
- stack inspection
- heap snapshots

Using these primitives, it is possible to build a hook system directly on top of the debugger. Importantly:

This approach does not rely on monkeypatching or modifying JavaScript objects. It operates entirely through debugger primitives.

Internally, this is implemented using:

- `Debugger.setBreakpoint` on function entry
- `Debugger.setBreakpoint` on return locations
- `Debugger.pause` + `Debugger.evaluateOnCallFrame`
- `Debugger.resume` / stepping primitives

No JavaScript objects are modified at runtime.

Example:

```
wb.hook("app.js:120:5", {
  onEnter(ctx) {
    ctx.log(ctx.arguments);
  },

  onLeave(ctx) {
    ctx.log(ctx.returnValue);
  }
});
```

Each hook is implemented using breakpoints at:

- function entry
- return points

When triggered, the handler runs inside the paused execution frame.

Context Design

Each handler receives a context (`ctx`) describing the runtime state:

```
ctx = {
  phase,
  stackTrace,
  variables,
  functionSource,
  returnValue
}
```

Handlers can:

- emit messages → `ctx.send(...)`
- modify variables → `ctx.setVariable(...)`

- override return values → `ctx.return(...)`
- evaluate expressions → `ctx.eval(...)`
- request continuation → `ctx.step*()` or `ctx.followReturn()`

It is worth noting that `ctx.eval()` executes code directly in the paused call frame of the target function, not inside an injected wrapper function. This allows evaluating expressions against the real runtime scope.

Stepping as a Programmable Primitive

Instead of manually pressing “Step Into” in DevTools, stepping becomes part of the API:

```
ctx.stepInto()
ctx.stepOver()
ctx.stepOut()
ctx.stepIntoAsync()
```

A new handler is introduced:

```
onStep(ctx, previousStep)
```

This is invoked after a step completes.

Crucially:

Stepping is no longer a UI action — it becomes a programmable loop.

Example:

```
onStep(ctx, prev) {
  if (ctx.variables.x > 10) return;
  ctx.stepInto();
}
```

This effectively turns the debugger into a scriptable execution engine.

Notably, Chrome DevTools does not expose `stepIntoAsync` in its UI.

This capability exists in CDP but is not directly available to users.

This model exposes it as a first-class primitive.

Following Execution Flows

A specialized primitive builds on top of stepping:

```
ctx.followReturn()
```

This requests:

- continue execution after return
- attempt to step into the next consumer
- trigger a new handler

Conceptually:

```
producer
↓
return
↓
follow
↓
consumer
```

followReturn is not a debugger primitive. It is a higher-level construct built on top of stepping + continuation matching.

It attempts to answer:

“Given this value, where is it used next?”

in a way that traditional debuggers cannot express directly.

Example: Tracing and patching an async Authentication Flow

```
async function authenticate(user, pass) {
  const token = await api.login(user, pass)
  const session = {
    token,
    isAdmin: utils.isAdmin(token)
  }
  saveSession(session)
}
```

Hook:

```
wb.hook("api.js:40:5", { // api.login location
  onLeave(ctx) {
    ctx.followReturn()
  },
  onReturnFollowed(ctx, previousStep){ // Here we are inside `authenticate` (line 1)
    ctx.log(ctx.variables.token)
  }
})
```

Execution:

```
api.login()
↓
returns token
↓
followReturn
↓
stepInto async continuation
↓
token is a live variable
```

This transforms debugging into **flow traversal**.

Patch the session:

We can hook the `authenticate` method directly:

```
wb.hook("main.js:102:32", { // `authenticate` location
  onLeave(ctx) {
    ctx.eval(`token.isAdmin=true`)
  }
})
```

How Continuation Tracking Works

Tracking the “next consumer” of a value is not directly supported by the runtime.

Continuation matching is therefore heuristic-based.

Async continuation tracking heuristics

`followReturn` relies on associating a step event back to its originating hook across an async boundary. The runtime does not provide this association directly — there is no CDP primitive that says "this pause is the continuation of that previous pause". There is no stable identifier for an async invocation exposed by the debugger (e.g. no call correlation ID).

The matching is therefore score-based, combining two signals.

Structural similarity. At every pause, the current stack trace is recorded and converted into a sequence of *trails* — one per frame, each expressed as `file:line:col`. To find the most likely continuation of a previous hook, the current trail sequence is compared against all previously recorded trail sequences. The score is the length of the longest common consecutive sequence between the two sequences. A longer shared sequence means the two pauses share more execution history, making it more likely they belong to the same logical flow.

Example:

Previous pause: `[ app.js:10:5 → auth.js:42:3 → api.js:88:10 ]`

Current pause: `[ app.js:10:5 → auth.js:42:3 → session.js:12:2 ]`

Common consecutive sequence: `[ app.js:10:5 → auth.js:42:3 ]`

Temporal proximity. Each pause is also assigned a step counter. Candidates with a smaller step distance receive a higher score. A continuation that arrived recently is more likely to be the correct one than one from much earlier in execution.

The two scores are combined, and the highest-scoring candidate is selected as the continuation.

This works well in practice for typical async flows. In heavily concurrent code with many interleaved async operations, the matching can become ambiguous — multiple candidates may share similar stack structure and arrive close together in time. In those cases the heuristic is best-effort.

This ambiguity is intrinsic to the runtime, not to the approach. Without explicit async call correlation exposed by the debugger, no tool operating at this level can resolve it deterministically. This is a fundamental limitation of debugger-level observability, not a limitation of this model.

Runtime Reality and Limitations

This approach is built on top of debugger primitives and inherits their constraints.

The async return value problem and continuation tracking heuristics are discussed in detail in the sections above. One further observation worth noting:

The debugger follows actual runtime execution — microtasks, scheduler, continuations — not the logical flow the developer expects.

This occasionally means that stepping and hook events fire in an order that feels surprising. The model is accurate. The intuition needs adjusting.

Performance vs Observability

This model intentionally trades runtime performance for observability.

The goal is not to minimize overhead, but to maximize visibility into execution.

For reverse engineering workflows, this trade-off is often acceptable.

Combining Runtime Domains

The approach becomes especially powerful when combined with:

- network interception
- runtime hooks
- memory inspection

Example workflow:

```
network response
↓
identify entry point
↓
hook function
↓
follow execution
↓
inspect objects
↓
patch logic
```

This enables end-to-end reverse engineering workflows inside the browser.

What This Enables

This model enables workflows that are impractical with traditional debugging:

- automatic flow traversal across async boundaries
- conditional execution steering
- dynamic patching of runtime logic
- correlation between network → runtime → memory

You stop navigating code. You start navigating execution.

Conclusion

Traditional debugging is manual.

Hooking improves visibility.

Event-driven debugging combines both:

```
runtime event
↓
handler
↓
debugger control
```

This turns the debugger into a **programmable execution engine**.

Instead of manually stepping through thousands of lines of code, it becomes possible to:

follow execution flows directly.

Resources

If this direction sounds interesting, this model is implemented in **Wirebrowser**

Hooks API are detailed here: <https://fcavallarin.github.io/wirebrowser/api/>

- **Frida-style JavaScript Hooks in the Browser Demo Video**

<https://www.youtube.com/watch?v=FtP8YFG9wSg>

`docs/index.md`

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layout: default title: Wirebrowser Documentation

Wirebrowser Documentation

Welcome to the documentation area for the Wirebrowser project. This section contains technical deep-dives and implementation notes for the core debugging and memory-analysis techniques.

This documentation covers two complementary approaches to JavaScript reverse engineering:

- **BDHS** — to identify where a value is created (origin tracing)

- **CDP Instrumentation** — to follow how that value propagates at runtime
-

□ **Breakpoint-Driven Heap Search (BDHS)**

BDHS is a temporal heap-analysis technique that performs step-out-based debugger pauses, captures a full heap snapshot at each stop, and searches each snapshot to identify where a value first appears or mutates inside modern, framework-heavy SPAs.

□ **Read the full technical writeup:** [Tracing JavaScript Value Origins with Breakpoint-Driven Heap Search \(BDHS\)](#)

⚙️ **CDP as a Runtime Instrumentation Engine**

This writeup explores how the Chrome DevTools Protocol can be used as a programmable runtime instrumentation engine — enabling function hooks, conditional stepping, and traversal of async execution flows without modifying the target application.

□ **Read the full technical writeup:** [CDP as a Runtime Instrumentation Engine - Hooks, Stepping, and Following Async Execution](#)

▶ **Automation Scripts API Reference**

The full API reference generated from the Wirebrowser type definitions is available here:

□ [Wirebrowser API Reference](#)

□ **Tutorials**

Step-by-step tutorials covering real-world runtime instrumentation workflows.

□ [Wirebrowser Tutorials](#)

□ **Additional Documentation**

More technical documents will be added here over time, including:

- Live Object Search internals
 - Structural Similarity Engine
 - Network-Memory correlation workflows
 - Architecture notes and reverse-engineering utilities
-

📄 Project Repository

For source code and installation instructions: → <https://github.com/fcavallarin/wirebrowser>

`docs/tutorials/Bypass-postMessage-checks.md`

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Bypassing postMessage origin checks with Wirebrowser (without modifying the source)

A common client-side security scenario is finding a `postMessage` handler but not being able to reach the interesting code path because of origin checks.

This guide shows how to bypass those checks at runtime using Wirebrowser.

Goal

Make the handler continue past its checks so we can observe or test the logic behind them.

Step 1: Find the target location

Before adding a hook, you need the location of the code you want to instrument.

The easiest way is to use the **Sources** panel in Wirebrowser:

1. Open the page in Wirebrowser
2. Go to **Sources**
3. Find the script containing the `postMessage` handler
4. Locate the function or line you want to instrument and place the cursor there
5. From the **Instrumentation** menu select **Create Hook at Cursor Position**

This will generate a hook template that you can edit to inject your logic.

Step 2: Where hooks live and how to run them

Hooks are defined inside **Node Scripts**, available in the **Automation** tab.

You can create them in two ways:

- manually, by creating a new script in **Automation → Node Scripts**
- from the **Sources → Instrumentation** menu (which generates a script automatically)

Once created, a hook is just part of a Node Script.

Hooks are activated by:

```
await WB.Node.Instrumentation.startHooks(pageId);
```

Running hooks

To activate your hooks:

1. Go to **Automation → Node Scripts**
2. Open your script
3. Click **Execute**

This will register all hooks defined in the script.

A single script can contain multiple hooks.

You can re-run the script anytime to re-register hooks or update their behavior.

Step 3: Stopping hooks

Hooks remain active until you explicitly stop them.

There are two ways to stop all active hooks:

Option 1: Stop hooks from the UI

In the top-right corner, there is a small bug icon .

- it turns green when the debugger is active
- each tab has its own debugger instance

Clicking the icon allows you to stop the debugger. When the debugger is stopped, all hooks are disarmed.

Option 2: Stop hooks programmatically

You can stop all hooks by running a script:

```
await WB.Node.Instrumentation.stopHooks()
```

This will remove all active hooks across all sessions.

Technique 1: Rebind the `event` parameter

```
window.addEventListener("message", function (event) {  
  if (event.origin !== "https://target.example") return;  
  handleMessage(event.data);  
});
```

Hook

```
WB.Node.Instrumentation.addHook({ file: "https://example.com/js/index.js", line: 16, col:
2 }, {
  onEnter(ctx) {
    event = {
      ...event,
      origin: "https://attacker.example"
    }
  }
});
```

This works because you replace the local `event` binding in the current frame.

Result

The handler now sees your modified origin and continues execution past the check.

Technique 2: Patch `allowedOrigin`

```
window.addEventListener("message", function (event) {
  const allowedOrigin = "https://target.example";
  if (event.origin !== allowedOrigin) return;
  handleMessage(event.data);
});
```

Hook

```
WB.Node.Instrumentation.addHook({ file: "https://example.com/js/index.js", line: 16, col:
2 }, {
  onEnter(ctx) {
    allowedOrigin = "https://attacker.example"
  }
});
```

This works without touching the `event` object at all.

Technique 3: Patch `allowlist`

```
const allowedOrigins = ["https://origin1.example", "https://origin2.example"];

if (!allowedOrigins.includes(event.origin)) return;
```

Hook

```
WB.Node.Instrumentation.addHook({ file: "https://example.com/js/index.js", line: 16, col:
2 }, {
  onEnter(ctx) {
    allowedOrigins.push("https://attacker.example");
  }
});
```

Technique 4: Hook helper function

```
if (!isAllowedOrigin(event.origin)) return;
```

Hook

Here we can place the hook inside `isAllowedOrigin` and override its return value.

```
WB.Node.Instrumentation.addHook({ file: "https://example.com/js/index.js", line: 2, col:
2 }, {
  onLeave(ctx) {
    ctx.return(true);
  }
});
```

This overrides the decision directly. This is often the cleanest approach when the check is encapsulated in a helper.

Technique 5: Patch at a precise location

In some cases, patching values at function entry is not enough.

For example:

```
function onMessage(event) {
  const parsed = JSON.parse(event.data);
  const origin = event.origin;

  if (origin !== allowedOrigin) return;

  handleMessage(parsed);
}
```

Here, `origin` is computed inside the function. While rebinding event at entry may still work, sometimes you want to patch the value exactly where it is used.

Hook at a specific location

```
WB.Node.Instrumentation.addHook({ file: "https://example.com/js/index.js", line: 2, col:
2 }, {
  at: [
    {
      location: "4:2", // line:col
      onHit(ctx){
        origin = allowedOrigin
      }
    }
  ]
});
```

This allows you to modify local variables or state at a precise execution point, rather than only at function boundaries.

When to use this

Use `at` when:

- the value you want to patch is created inside the function
- you want to observe or modify intermediate state
- function-level hooks (`onEnter`) are too early or too broad

Notes

- Run hooks at function entry to ensure values are patched before checks execute
- Ensure your fake `event` includes the fields the handler actually uses (`data` , `source` , etc.)
- Prefer patching application logic over browser-native objects when possible

Summary

- Rebind `event` to control handler input
- Patch variables and allowlists used in checks
- Override helper return values when logic is encapsulated

The key idea: you don't need to modify the browser event — you only need to control what the handler reads.

`docs/tutorials/index.md`

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Tutorials

This section contains practical, step-by-step guides for using Wirebrowser in real-world scenarios.

Wirebrowser is a runtime instrumentation platform built on the Chrome DevTools Protocol (CDP).

Basics

- **Create Hooks and patch application behavior**

[Bypass postMessage checks](#)

Notes

- Wirebrowser operates at the debugger level (CDP), not at the source level.
 - No monkeypatching or source rewriting is required.
 - Some behaviors (especially async) follow debugger semantics, not JavaScript semantics.
-

Archived from <https://github.com/fcavallarin/wirebrowser>. Rendered offline from the local Markdown copy.